



National University

of Computer & Emerging
Sciences Islamabad

Department of DS/AI

Calculus and Analytical Geometry

Assignment#01

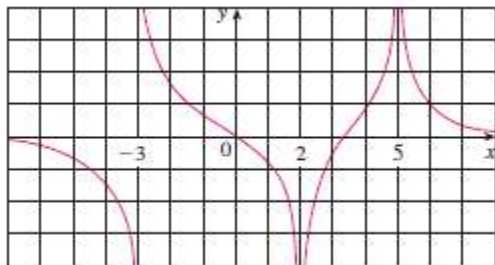
Semester: Fall 2026

Due Date: 15-09-2026

Course Instructor: Arif ,Izza Shafique

Total Marks: 130

1. For the functions whose graph is shown, state the following (10)
- a) $\lim_{x \rightarrow -7} f(x)$ b) $\lim_{x \rightarrow -3} f(x)$ c) $\lim_{x \rightarrow 0} f(x)$ d) $\lim_{x \rightarrow 6^-} f(x)$ e) $\lim_{x \rightarrow 6^+} f(x)$
- f) The equations of the vertical asymptotes



2. Sketch the graph of the function f with the specified properties (10)
- i) The domain of f is $[-1, 1]$
 - ii) $f(-1) = f(0) = f(1) = 0$
 - iii) $\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 1} f(x) = 1$.
3. Find the limits if possible (10)
- i) $\lim_{t \rightarrow 9} \frac{9-t}{\sqrt{3}-t}$
 - ii) $\lim_{y \rightarrow -\infty} \frac{2-y}{\sqrt{7+6y^2}}$

$$\text{iii) } \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\text{iv) } \lim_{x \rightarrow \pi} \cos^2(x - \tan x)$$

4. Use the squeeze theorem to show that $\lim_{x \rightarrow 0} (x^2 \cos 2\pi x) = 0$. Illustrate by graphing the functions $f(x) = -x^2$, $g(x) = x^2 \cos 2\pi x$ and $h(x) = x^2$ on the same screen. (10)
5. Suppose that $f(x)$ and $g(x)$ are polynomials in x . Can the graph of $\frac{f(x)}{g(x)}$ have any asymptote if $g(x) \neq 0$? Give reasons of your answer. (10)
6. Suppose that a function is continuous everywhere and that $f(-2) = 3$, $f(-1) = -1$, $f(0) = -4$, $f(1) = 1$ and $f(2) = 5$. Does intermediate value theorem guarantee that f has a root on the following intervals? (10)
- a) $[-2, -1]$ b) $[-1, 0]$ c) $[0, 2]$
7. Find the values of m and k if possible that will make the function continuous everywhere?

$$\begin{cases} x^2 + 5 & x > 2 \\ m(x + 1) + k & -1 < x \leq 2 \\ 2x^3 + x + 7 & x \leq -1 \end{cases}$$

(10)

8. Identify the existence of Horizontal, Vertical or Oblique asymptote if it exists? (10)

i) $\frac{2x^3 + 7}{x^3 - x^2 + x + 7}$

ii) $\frac{3x^7 + 5x^2 - 1}{6x^3 - 7x + 3}$

9. Sketch the graph of the function if $f(0) = 0$, $\lim_{x \rightarrow \pm\infty} f(x) = 0$, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) = \infty$, $\lim_{x \rightarrow 1^+} f(x) = -\infty$ and $\lim_{x \rightarrow -1^-} f(x) = -\infty$. (10)
10. Graph the function and find the zeros and the domain and the range: (10)

$$f(x) = \sqrt[3]{|9 - x^2|} - 1$$

11. Find the equation of the new curve C obtained when the graph of the curve C with the equation $x^2 - 12x + 3y = 1$ is reflected in the origin. (10)
12. The function $f(x) = \sqrt{x}$ is reflected across the y-axis and then stretched vertically by a factor of 2 to create $g(x)$.
- (a) Write the equation for $g(x)$.

(b) Determine the domain and range of $g(x)$.

(c) Verify using Python.

(10)

13. Find all the possible asymptotes of given function and verify by using Python

$$f(x) = \frac{2x+1}{x-3}.$$

(10)